

TD1 The **Fourier transform (FT)**

Context :

The **Fourier transform (FT)** is an integral transform that converts a function into a form that describes the frequencies present in the original function. The output of the transform is a complex-valued function of frequency.

But we will only work with the real part of this complex value.

The Fourier Transform is sometimes called the frequency domain representation of the original function.

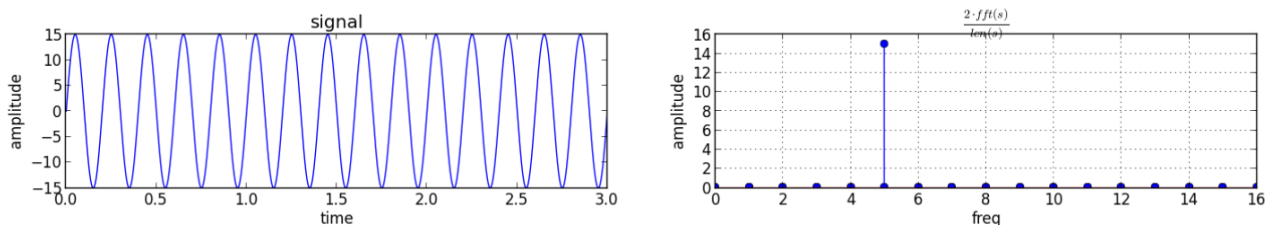
Besides the original Fourier Transform includes :

- Fourier series . The latter is routinely employed to handle periodic functions.
- The Discrete Fourier transform (DFT) is employed to handle sampled functions
- The Fast Fourier transform (FFT) is an algorithm for computing the DFT.

For exemple : The FT of sinus

Sine wave (left column) and his respective frequency domain representations (right column)

The frequency plot has exactly one peak at x=5 (frequency), y=15 (amplitude)

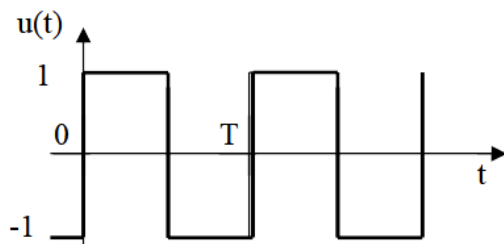


Exercise #1: Fourier series decomposition

The Fourier series is an example of a trigonometric series

$$A_0 + \sum_{n=1}^{\infty} A_n \cos nx + B_n \sin nx,$$

We consider the signal $u(t)$ in time domain with time period $T=1\text{ms}$ described below



A rectangular signal is the addition of various sinusoidal oscillation. Its Fourier series decomposition is :

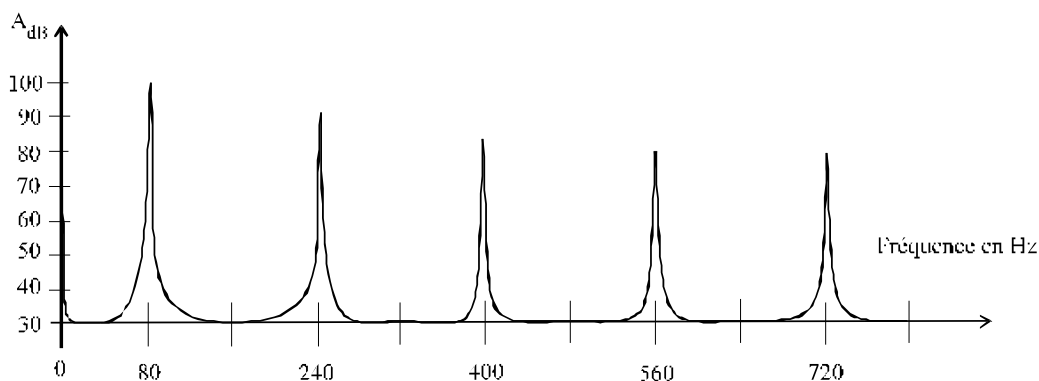
$$u(t) = \frac{4}{\pi} \left[\sin \omega \cdot t + \frac{\sin 3\omega \cdot t}{3} + \frac{\sin 5\omega \cdot t}{5} + \dots \right]$$

The greater the number of these components, the closer the signal approaches the ideal rectangular pulse

=>Plot the frequency domain representation of u(t)

Exercise #2: the frequency domain representation

The frequency analysis of a periodic signal is described **below**:



1°) What is the frequency of the fundamental ?

2°) The time period of the signal is determined by the frequency of the fundamental

- What is the period T of this signal?
- What are the amplitudes of even harmonics
- What are the amplitudes of odd harmonics

2°) Gain is a measure of the ability of a two-port circuit (often a filter) to increase the amplitude of a signal from the input to the output port. $A_{dB} = 20 \log_{10} \left(\frac{V}{V_{ref}} \right)$.

For exemple if $V=10\text{mV}$ and $V_{REF}=1\text{mV}$ then $A_{dB}=20\text{dB}$

A gain greater than zero dB means amplification of the circuit

The term gain is ambiguous and only used with current, voltage. We will use « power gain » with power : $A_{dB} = 10 \log_{10} \left(\frac{P}{P_{ref}} \right)$.

For $V_{ref} = 1 \mu\text{V}$ and $A_{dB} = 20 \log_{10} \left(\frac{V}{V_{ref}} \right)$, calculate the amplitude of the fundamental, noted VF

Exercise #3:

A Master IOM student browses the Internet in 3G (UMTS) with his phone. 3G communications can use 2 frequency bands in FDD "Frequency Division Duplexing" mode (1 frequency band for the uplink and 1 for the downlink). The 1900-2100 MHz band is reserved for 3G and it is also possible to use the GSM 900 or 1800 MHz bands. In the latter case the operator chooses whether he wants to do 2G (GSM) or 3G (UMTS) in the frequency band allocated to it.

We give the distribution of frequencies in the 900MHz band.

Looking at the DSP of the GSM 900 below, represent the different operator bands on the spectrum .

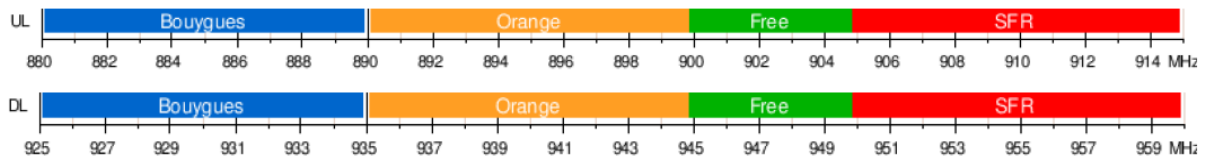


Figure 3 : Band of 900MHz (shared with GSM)

Question :

- 1°) Looking at the DSP of the GSM 900 downband below, represent on the spectrum below, the different operator bands.
 - 2°) What is the SPAN of this frequency domain representation?
 - 3°) What is the gain of the noise?
 - 4°) What is the gain of the greatest signal?
 - 5°) What is the value of the NSR (Noise signal ratio)of the greatest signal?
 - 6°) The resolution bandwidth (RBW) is the smallest step between 2 values of frequency. Ideally, you would like to set this bandwidth as narrow as possible, as that would give you the finest frequency resolution.
- How many acquisition points does this spectrum contain?

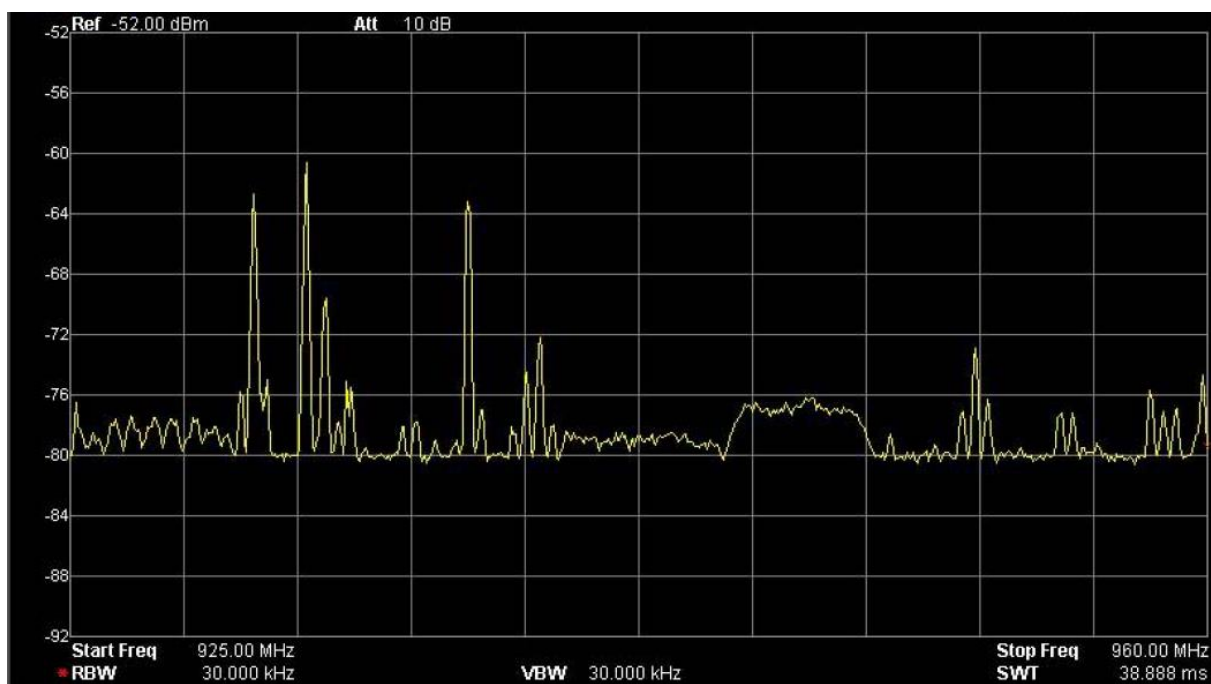


Figure 4 : FFT of the downlink band

Context :

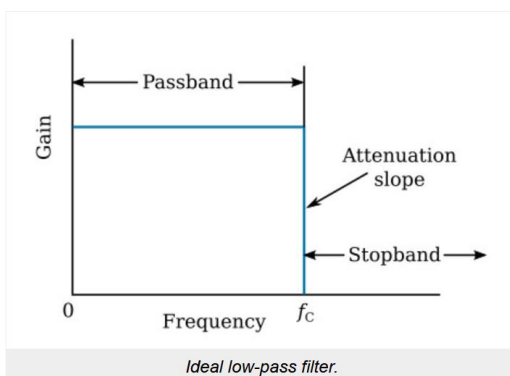
A filter is a device or process that removes some unwanted components from a signal.

A filter may be required to pass or attenuate input signals so that the output amplitude-versus-frequency curve has some desired shape.

The equations that describe output-versus-input relationships of filters can be written directly in a mathematical form called a transfer function=> $H(\omega) = \underline{V}_{out}/\underline{V}_{in}$



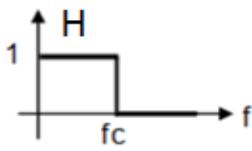
When a voltage is applied to the filter of figure below, the complex transfer function is $H(\omega)$. The magnitude of the complex transfer function $|H(j\omega)|$ is then referred to as amplitude response, while the phase of the complex transfer function is called phase response.



The frequency response can be classified into a number of different bandforms :

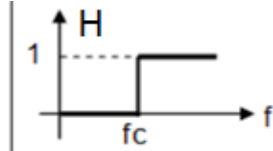
- Low-pass filter – low frequencies are passed, high frequencies are attenuated.
- High-pass filter – high frequencies are passed, low frequencies are attenuated.
- Band-pass filter – only frequencies in a frequency band are passed.

Cutoff frequency is the frequency beyond which the filter will not pass signals. It is usually measured at a specific attenuation such as 3 dB



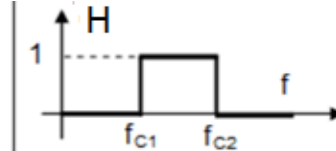
Low-pass filter

The passband is $[0, f_c]$



High-pass filter

The passband is $[f_c, \infty]$



Band-pass filter

The passband is $[f_{c1}, f_{c2}]$

Exercise n° 1 :



PSD=power spectral density

We give the spectral occupancy spectrum (PSD: power spectral density) of a private phone line. This spectral representation will be called $X(f)$

We have 2 filters:

an ideal low pass filter of $f_{c1}=10\text{kHz}$

and an ideal high pass filter of $f_{c2}=15\text{kHz}$.

1. Draw the ideal transfer function $H_1(f)$ of the low pass filter
2. Draw the ideal transfer function $H_2(f)$ of the high pass filter
3. This line is filtered by the low pass filter previously studied. Draw the filtered transfer function: $X_1(f)=X(f).H_1(f)$
4. This line is filtered by the high pass filter previously studied. Draw the filtered transfer function: $X_2(f)=X(f).H_2(f)$

Exercise #5: Transfert Function of raised-cos filter

A raised-cos filter is a filter used, in Gnuradio's PSK modulation blocks.

Its transfer function $H(\omega)$ has the following characteristic:

- $H(\omega)=1$ for $\omega < \omega_c.(1-\alpha)$ with α = roll-off factor
- $H(\omega)=0$ for $\omega > \omega_c.(1+\alpha)$

$$\bullet \quad H(\omega)=\sqrt{\frac{1+\cos\left(\frac{\pi\cdot(\omega-\omega_c\cdot(1-\alpha))}{2\cdot\alpha\cdot\omega_c}\right)}{2}}$$

1. Plot $H(\omega)$ for $\omega_c=1000$ rad/s and $\alpha=0.35$ and for the following pulsations: $\omega=100$, 500, 1000, 1500, 2000, 5000 rad/s
2. Why it's called raised-cos?

Context :

For an exact calculation of the frequency spectrum of an input signal, an infinite period of observation would be required.

In order to respect the Fourier Transform law ($X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j2\pi ft} \cdot dt$),

The signal amplitude should be known at every point in time.

The result of this calculation would be a continuous spectrum, so the frequency resolution would be unlimited (It always has components frequency between 2 components)

It is obvious, that such exact calculations are not possible in practice.

Given certain prerequisites, the spectrum can nevertheless be determined with sufficient accuracy.

In practice, the signal to be analysed has to be sampled by an analog digital converter.

By the way of sampling, the continuous input signal is converted into a discrete time signal (see figure 1)

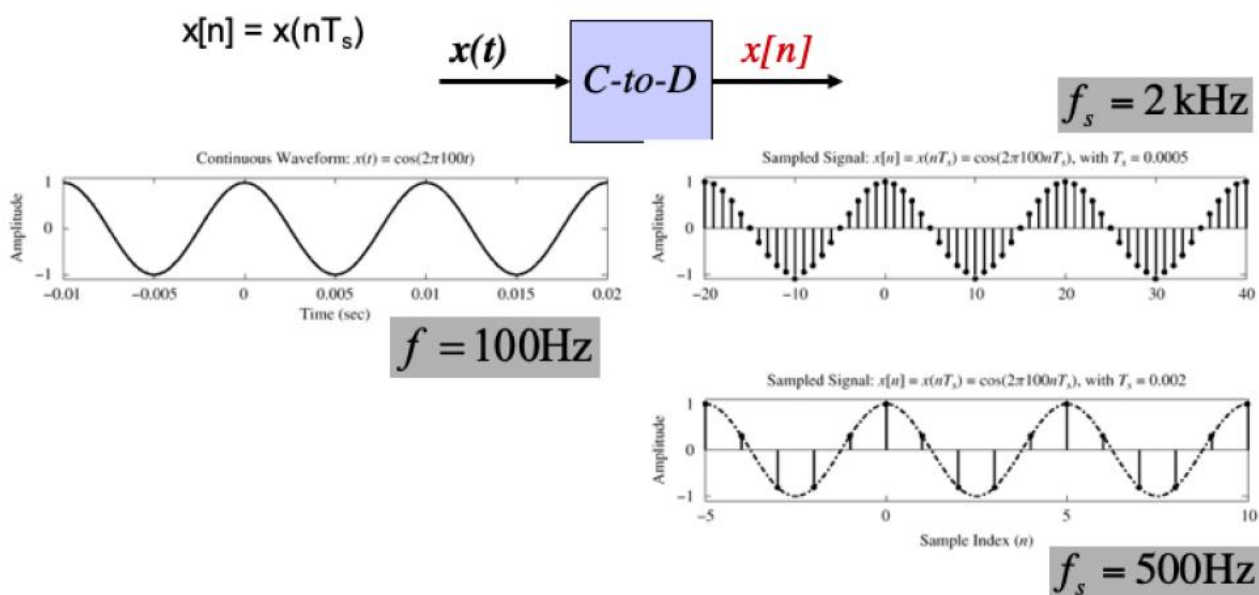


Figure 1 : sampled sinus signal

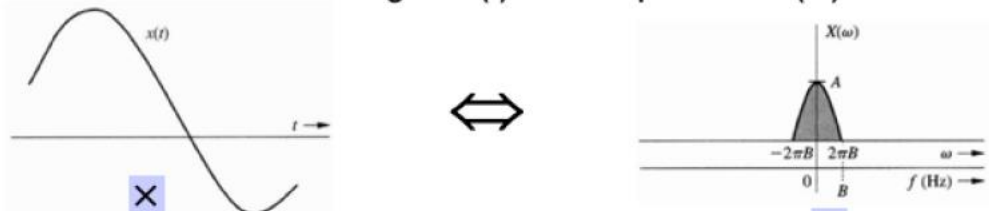
We take a sample of the signal $x(t)$ at regular interval of period T_s

Sampling done properly, will not destroy information. That is, you can always get back the original signal with no loss information.

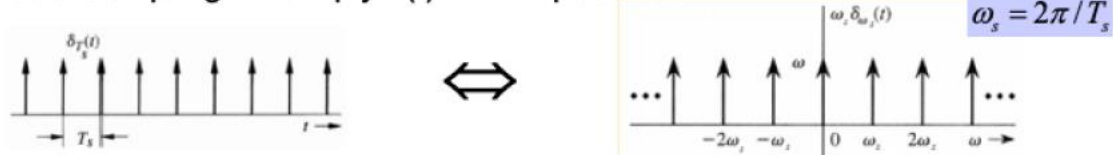
That means you have retained enough samples of the original signals.

MULTIPLYING two signals in the time domain is equivalent to CONVOLUTING their spectral in the frequency domain (see figure 2)

- Consider a bandlimited signal $x(t)$ and its spectrum $X(\omega)$:



- Ideal sampling = multiply $x(t)$ with impulse train :



- Therefore the sampled signal has a spectrum (convolution):

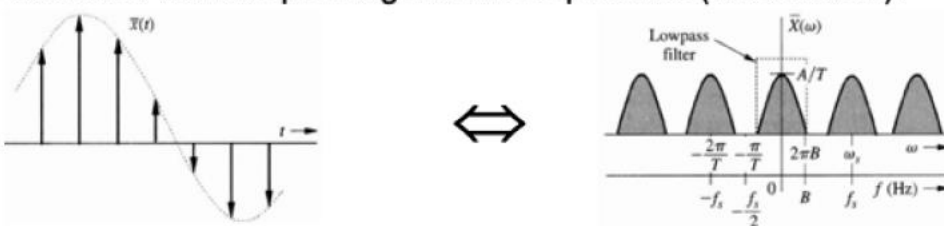


Figure 2 : convolution

The effect of convoluting the spectrum of the sampling impulse train with the original spectrum $X(\omega)$ is to replicate the shape of $X(\omega)$ wherever there is an impulse in the frequency spectrum

Sampling theorem (sometimes also known as Shannon theorem or Nyquist theorem)

If the original signal has a MAXIMUM frequency component at $f_{\text{BIN_MAX}}$, then we MUST sample at $2.f_{\text{BIN_MAX}}$ or higher in order NOT to lose information.

Simply put : Sampling frequency $f_s > 2 f_{\text{BIN_MAX}}$

If you ignore this rule, you will corrupt the sampled signal by introducing spurious signals through a phenomenon known as « aliasing »

The bandwidth of the input signal must be limited or else the higher signal frequency will cause aliasing effects due to sampling. (see figure 3)

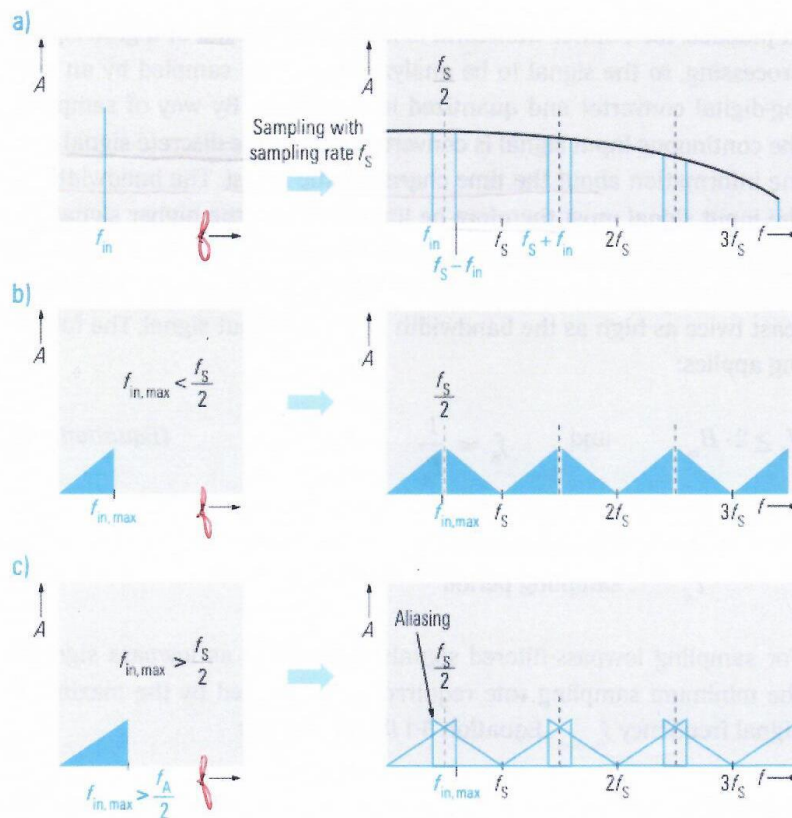
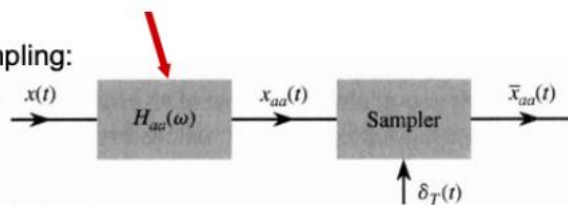


Figure 3 : impact of aliasing

The anti-aliasing filter must be applied before the sampling step (see figure 4)

Anti-aliasing filter

- ◆ Apply a lowpass filter before sampling:



- ◆ Now reconstruction can be done without distortion or corruption to lower frequencies:

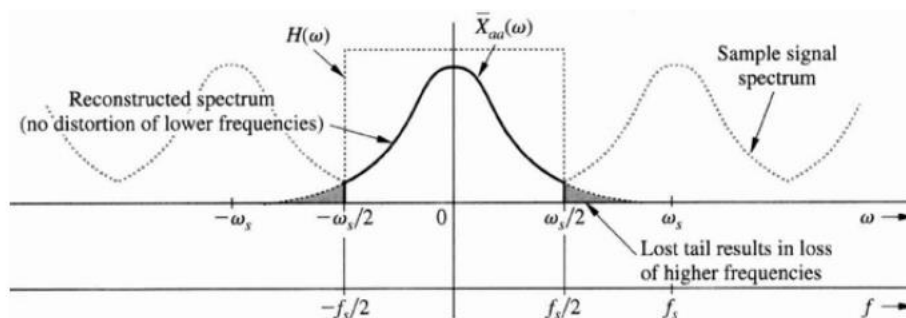


Figure 4: anti aliasing filter

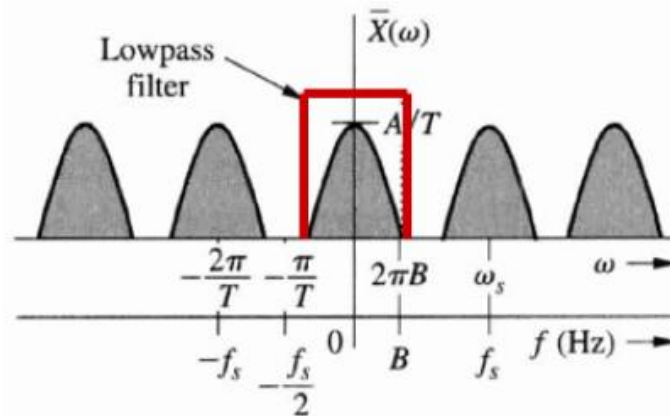
We must always apply a lowpass filter to the continuous time signal BEFORE sampling is performed.

This cuts off the frequency beyond $f_s/2$

Reconstruction :

The original signal $x(t)$ is represented by the main shape in the spectral domain around frequency=0. (see figure 5)

- ◆ Therefore, to reconstruct the original signal $x(t)$, we can use an ideal lowpass filter on the sampled spectrum:



- ◆ This is only possible if the shaded parts do not overlap. This means that f_s must be more than TWICE that of B .

Figure 5 : Reconstruction filter

If we apply a filter to the sampled signal and extract the main spectral components within the RED rectangle in the frequency domain, we can recover the original signal with no loss information. The filtering process is a lowpassfilter where we only retain frequency components within $\pm B$ Hz and remove all other high frequency components. This is known as « reconstruction »

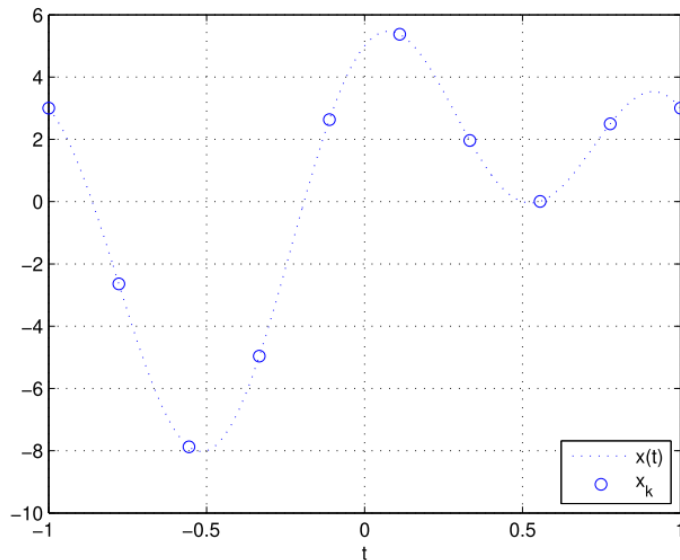
Exercise 1: Aliasing

We Consider two sinusoidal signals, one $x(t)$ at 1Hz and another $y(t)$ at 6Hz

1. Plot in the same time representation $x(t)$ et $y(t)$
2. Plot $x^*(t)$ sampled signal of $x(t)$. The signal will be sampled at a rate of 5Hz
3. Plot $y^*(t)$ sampled signal of $y(t)$.The signal will be sampled at a rate of 5Hz
4. Are sinewaves distinguishable after sampling ?
5. Which signal respects the Nyquist theorem ?

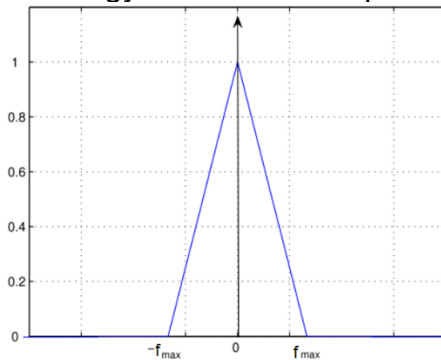
Exercise 2: Sampling

We consider the samples x_k of a sampled signal. What is the sampling frequency?

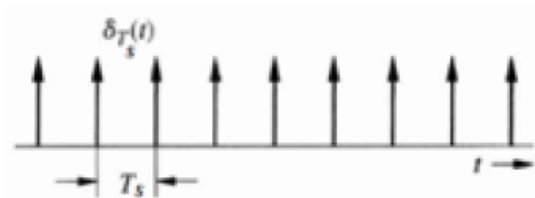


Exercise #3: FOLDED SPECTRUM

We consider the spectrum $X(f)$ of the signal $x(t)$.
Its energy is limited to frequency of $[-f_{\max}, f_{\max}]$.



We consider the impulse train time representation $i(t) = \sum_{n=-\infty}^{\infty} \delta(t - nTs)$

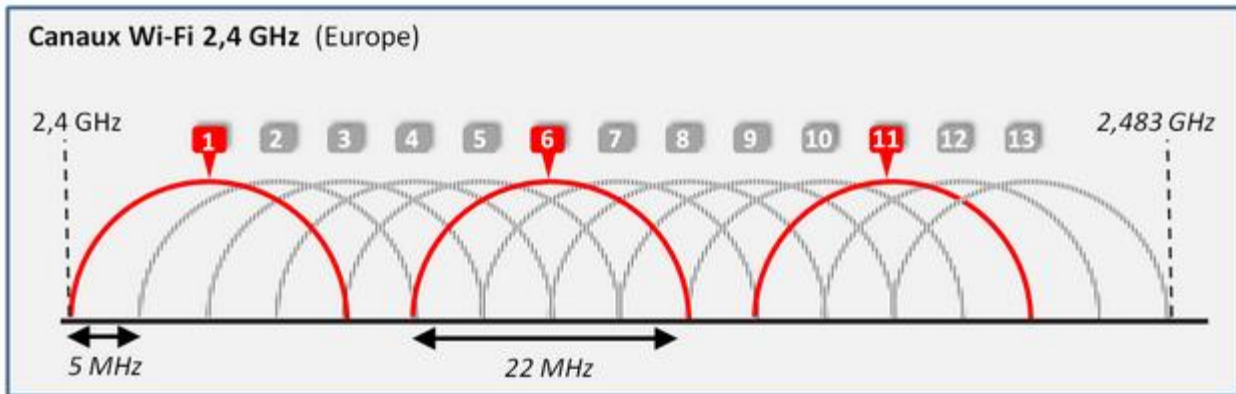


1. Plot the spectrum $I(f)$ of the signal $i(t)$.
2. Build the sampled spectrum $X^*(f)$ corresponding to $x(t)^*$ (you need to relocate the shape of $X(f)$ wherever there is an impulse in the spectrum $I(f)$).
 $x(t)^* = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nTs) \Rightarrow X^*(v) = X(v) * \sum_{n=-\infty}^{\infty} \delta(v - n/Ts)$. Its is the convolution relation
3. we are going to study spectrum aliasing .
 - a) First case : Plot the spectrum $X^*(f)$ for $f_{\max} = 1000\text{Hz}$ et $1/Ts = 3000\text{Hz}$. Did you respect Nyquist theorem ?
 - b) Second case : Plot the spectrum $X^*(f)$ for $f_{\max} = 1000\text{Hz}$ et $1/Ts = 1500\text{Hz}$. Did you respect Nyquist theorem ?
 - c) In which case can we reconstruct the signal?

Exercise 4 : Sampled frequency

When we talk about 2.4 GHz frequencies, there is one frequency band divided into several 20 MHz channels. When deploying a wireless architecture, the choice of different channels is essential to minimize the risk of interference.

We want to use Gnuradio to acquire the wifi band. What sampling frequency should we choose?



TD4 Windowing in **Fast Fourier transform (FFT)**

Context :

Only a limited number N of samples is used for calculation in the Fast Fourier Transform (see figure 1). This process is called **WINDOWING**

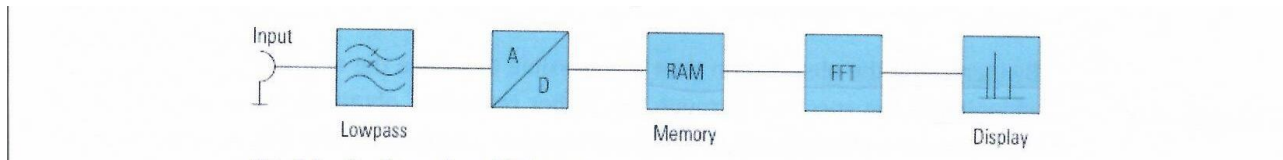


Figure 1 : Chain to display a FFT

The input signal is multiplied with a specific window function before or after sampling in the time domain.

In figure 2, a rectangular window is used.

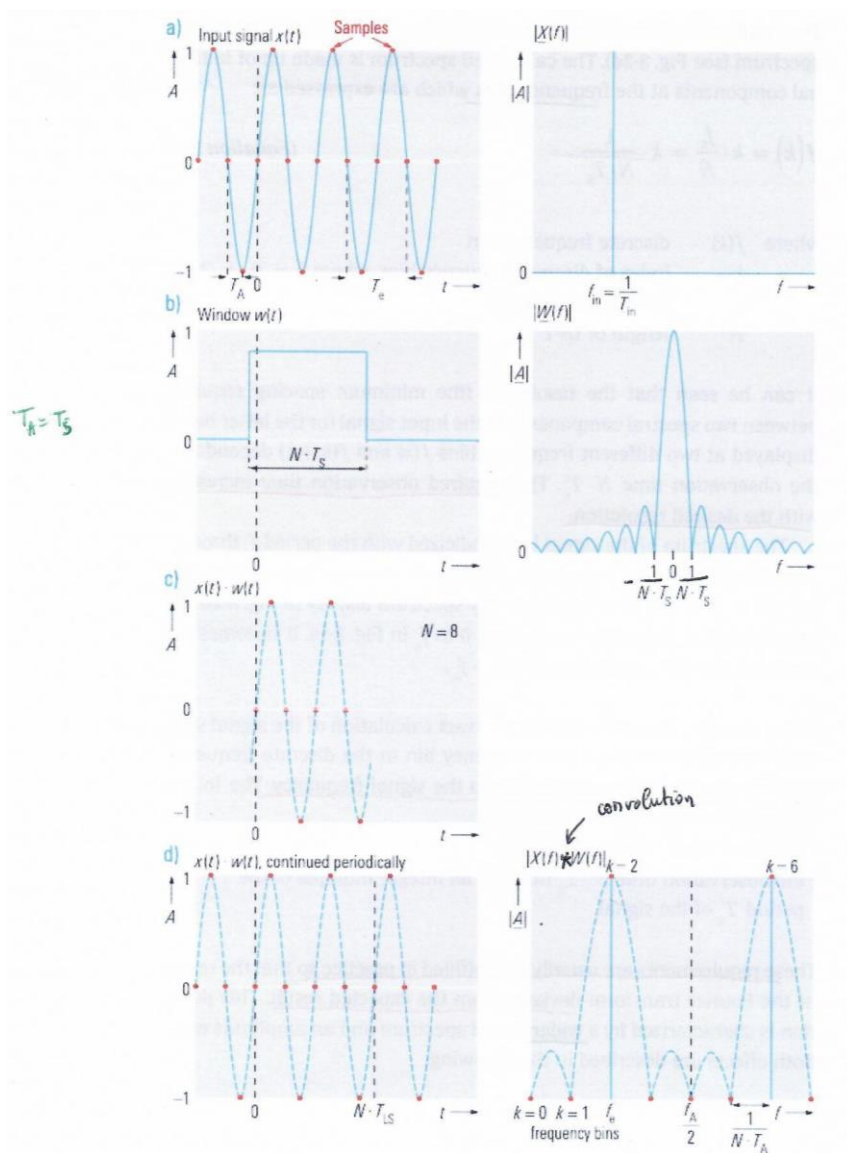


Figure 2:
DFT with periodic
input signal.
Observation time is an
integer multiple of the
period of the input
signal

It is a particular case :

- The signal is periodic (T_0)
- The observation time $N \cdot T_S$ must be an integer multiple of the period T_0 of the signal

These requirements are usually not fulfilled in practice so that the Fourier Transform deviates from the expected result. This deviation is characterized by a wider signal spectrum and an amplitude error. Both effects are described in figure 3

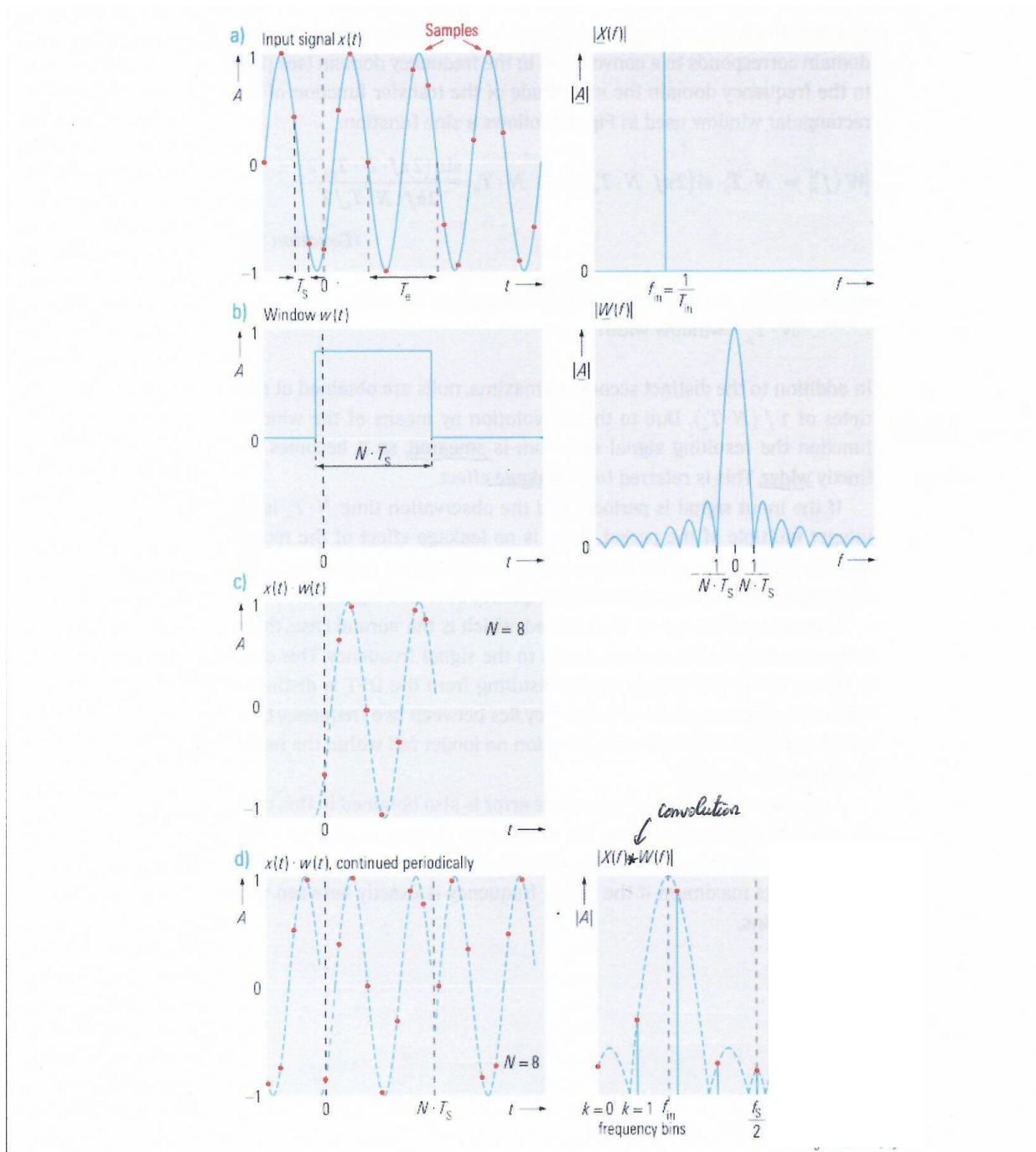


Figure 3: DFT with periodic input signal. Observation time is not an integer multiple of the period of the input signal.

PS : In the frequency domain, the magnitude of the transfert function of the rectangular window follows a function named SINUSCARDINAL

The two effect can be reduced by using optimised windowing instead of rectangular window . In figure 4, the leakage effect is reduced by a HANN WINDOW (but an magnitude error will appear)

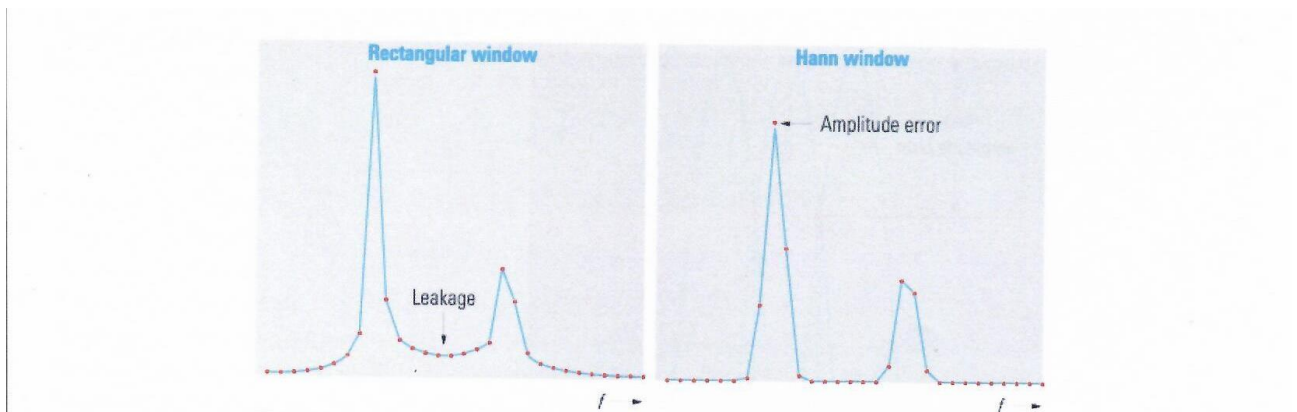
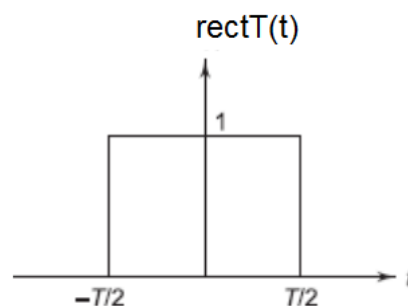


Figure 4: leakage effect when using rectangular window or Hann window

A FLAT-TOP WINDOWS is used to obtain a high level accuracy . Indeed, the maximum level error of this windowing function is as small as 0.05dB. A drawback is its relatively wide main lobe which reduces the frequency resolution

Exercise #1: cardinal sine function

The gate function $\text{rectT}(t)$

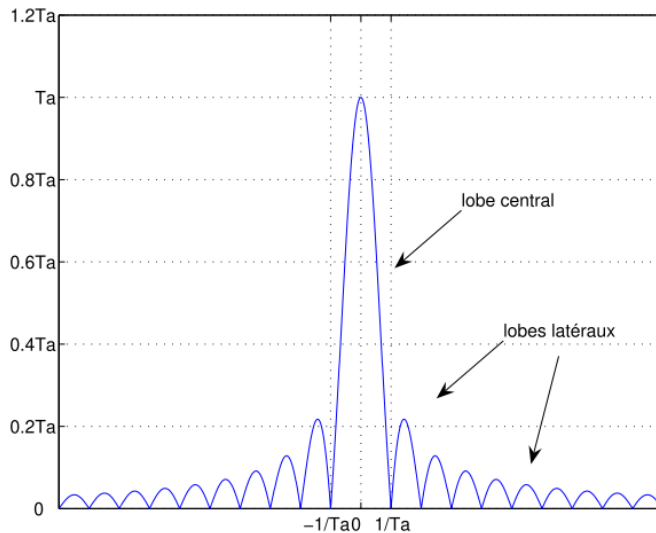


Mathematically, it can be written as

$$\text{rectT}(t) = \left. \begin{array}{l} 0 \text{ for } t < -\frac{T}{2} \\ 1 \text{ for } -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0 \text{ for } t > \frac{T}{2} \end{array} \right\}$$

The cardinal sine function is the Fourier Transform of $\text{rectT}_A(t)$

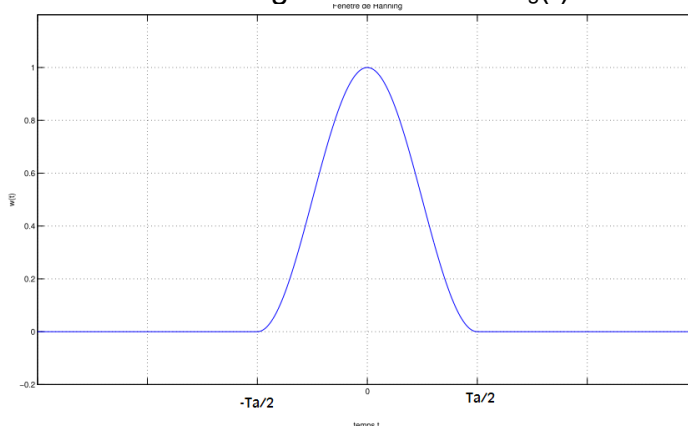
so $TF(\text{rect}T_A(t)) = \text{sinc}(\pi f)$. The magnitude of this TF is the spectrum below:



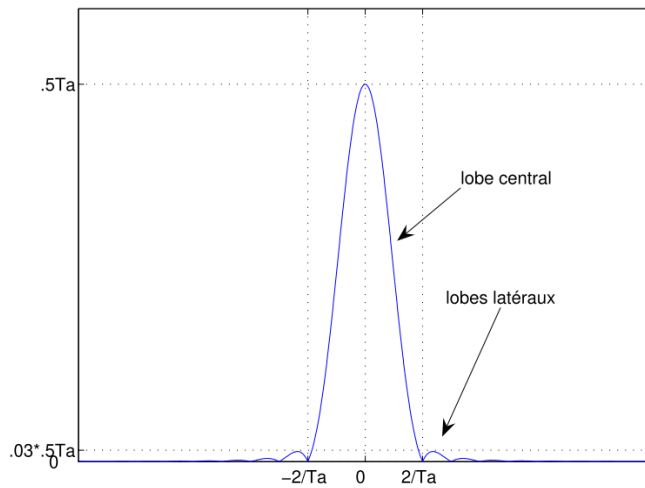
1. Determine the frequencies for which the side lobes are maximum.
2. Determine the frequencies for which the side lobes are zero.
3. Determine the width of the central lobe.
4. Determine the bandwidth necessary for this TF, if we consider all lobes $> 3\%$ of the main lobe.

Exercice 2 : Hanning function

Here is the Hanning function: $w_{\text{Hanning}}(t)$



- a) Describe the difference between a gate function and a Hanning function.
- b) We give below the TF of the Hanning function. Determine the bandwidth necessary for this TF, if we consider all lobes $> 3\%$ of the main lobe.
- c) Determine the width of the central lobe
- d) Compare these results between the TF of the gate function and the Hanning function.



Exercise #3:

We wish to acquire with Gnuradio a Lora signal (Long Range) whose bandwidth is 250kHz.

This signal is FM modulated. The spectral resolution should be smaller than 1kHz.

Determine :

1. the sampling frequency,
2. the number of points
3. and the type of window (rectangular, Hanning, Flat-top)